CAPS-SearchSpace

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1 Dataset

Table 1: Dataset details used in experiments.

Dataset	# Rows (Instances)	# Columns (Features + Target)
philippine	5,832	309
jasmine	2,984	145
jannis	83,733	55
helena	65,196	28
fabert	8,237	801
dilbert	10,000	2,001
robert	10,000	7,201
albert	425,240	79
digits	1,797	65
dionis	416,188	61

2 Search Space Option 1: Detailed Table

Table 2: AutoML Search Space Configuration

Preprocessors (14)	**Hyperparameters**
Binarizer	threshold $\in [0.0, 1.0]$ (step 0.05)
FastICA	tol $\in [0.0, 1.0]$ (step 0.05)
FeatureAgglomeration	linkage $\in$ {ward, complete, average}, affinity $\in$ {euclidean, l1, l2, manhattan, cosine}
MaxAbsScaler, MinMaxScaler	No hyperparameters
RobustScaler, StandardScaler, ZeroCount	No hyperparameters
Normalizer	norm $\in$ {l1, l2, max}
Nystroem	kernel $\in$ {rbf, cosine, chi2, ...}, gamma $\in [0.0, 1.0]$, n_components $\in [1, 11]$
PCA	svd_solver = randomized, iterated_power $\in [1, 11]$
PolynomialFeatures	degree = 2, include_bias = False
RBFSampler	gamma $\in [0.0, 1.0]$ (step 0.05)
OneHotEncoder	minimum_fraction $\in \{0.05, 0.1, 0.15, 0.2, 0.25\}$
Feature Selectors (5)	**Hyperparameters**
SelectFwe	alpha $\in [0, 0.05]$ (step 0.001), score_func = f_classif
SelectPercentile	percentile $\in [1, 100]$, score_func = f_classif
VarianceThreshold	threshold $\in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}, 10^{-1}, 2 \times 10^{-1}\}$
RFE	step $\in [0.05, 1.0]$, estimator = ExtraTreesClassifier
SelectFromModel	threshold $\in [0, 1.0]$, estimator = ExtraTreesClassifier
Classifiers (13)	**Key Hyperparameters**
GaussianNB	No hyperparameters
BernoulliNB, MultinomialNB	alpha $\in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$, fit_prior $\in$ {True, False}
DecisionTreeClassifier	criterion $\in$ {gini, entropy}, max_depth $\in [1, 11]$, min_samples_split $\in [2, 21]$
RandomForest, ExtraTrees	n_estimators = 100, max_features $\in [0.05, 1.0]$, bootstrap $\in$ {True, False}
GradientBoosting	n_estimators = 100, learning_rate $\in \{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$, max_depth $\in [1, 11]$
KNeighborsClassifier	n_neighbors $\in [1, 101]$, weights $\in$ {uniform, distance}, p $\in \{1, 2\}$
LinearSVC	penalty $\in$ {l1, l2}, C $\in \{10^{-4}, ..., 25\}$, dual $\in$ {True, False}
LogisticRegression	penalty $\in$ {l1, l2}, C $\in \{10^{-4}, ..., 25\}$, dual $\in$ {True, False}
XGBClassifier	n_estimators = 100, max_depth $\in [1, 11]$, learning_rate $\in \{10^{-3}, ..., 1\}$
SGDClassifier	loss $\in$ {log, hinge, ...}, penalty = elasticnet, alpha $\in \{0, 0.01, 0.001\}$
MLPClassifier	alpha $\in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, learning_rate_init $\in \{10^{-3}, ..., 1\}$

3 Search Space Option 2: Compact Summary Table

Table 3: AutoML Search Space Summary

Component	**Count**	**Categories**	**Key Parameter Ranges**
Preprocessors	14	Scalers (6), Transformers (5), Decomposition (2), Encoding (1)	Continuous: $[0, 1]$ Discrete: varies
Feature Selectors	5	Statistical (3), Model-based (2)	α: $[0, 0.05]$, percentile: $[1, 100]$, threshold: $[0, 1]$
Classifiers	13	Tree-based (4), Linear (3), Probabilistic (3), Neural (1), Other (2)	n_estimators: 100, max_depth: $[1, 11]$, C: $[10^{-4}, 25]$
Total Operators	**32**		**Experiments: 3 random seeds**

4 Search Space Option 3: Split into Two Smaller Tables

Table 4: Search Space Operators

Type (Count)	**Operators**
Preprocessors (14)	Binarizer, FastICA, FeatureAgglomeration, MaxAbsScaler, MinMaxScaler, Normalizer, Nystroem, PCA, PolynomialFeatures, RBFSampler, RobustScaler, StandardScaler, ZeroCount, OneHotEncoder
Feature Selectors (5)	SelectFwe, SelectPercentile, VarianceThreshold, RFE, SelectFromModel
Classifiers (13)	GaussianNB, BernoulliNB, MultinomialNB, DecisionTreeClassifier, ExtraTreesClassifier, RandomForestClassifier, GradientBoostingClassifier, KNeighborsClassifier, LinearSVC, LogisticRegression, XGBClassifier, SGDClassifier, MLPClassifier

Table 5: Key Hyperparameter Ranges

Parameter Type	**Common Ranges**
Tree depth	$[1, 11]$
Number of estimators	100 (fixed)
Regularization (C)	$\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1, 5, 10, 15, 20, 25\}$
Learning rate	$\{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$
Alpha (NB, NN)	$\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$
Min samples split/leaf	$[1, 21]$
Max features	$[0.05, 1.0]$ (step 0.05)
K neighbors	$[1, 101]$
Percentile	$[1, 100]$
Threshold	Context-dependent